Lesson 1: Squares, Square Roots and Triangle Basics

Assignment

1. Find each square. Use your calculator for (g) to (l).

(a) 5^2 _____ (b) 12^2 _____ (c) 1^2 _____ (d) 10^2 _____ (e) 40^2 _____ (f) 0.5^2 _____

(g) 17^2 _____ (h) 3.2^2 _____ (i) 0.8^2 _____ (j) 26^2 _____ (k) 1.25^2 _____ (l) 7.5^2 _____

2. Circle the numbers that are perfect squares.

16 24 49 90 121 150 196 1000

Pick one number that is *not* a perfect square and explain how you know.

3. Find the area of each square.

(a) A square sticky note that is 7.5 cm on each side.

(b) A square deck that is 14 ft on each side.

(c) A square boxing ring that is 6.1 m on each side.

4. Find each square root without a calculator.

(a) $\sqrt{49}$ _____ (b) $\sqrt{144}$ _____ (c) $\sqrt{4}$ _____ (d) $\sqrt{400}$ _____ (e) $\sqrt{121}$ _____ (f) $\sqrt{196}$ _____

5. Use your calculator. Round each square root to the nearest tenth.

(a) $\sqrt{20}$ _____ (b) $\sqrt{75}$ _____ (c) $\sqrt{200}$ _____ (d) $\sqrt{3}$ _____ (e) $\sqrt{12.5}$ _____

(f) Check your answer to (b) by squaring it. Is it close to 75?

6. Estimate each square root without a calculator. Then check with your calculator (nearest tenth).

Number	Between which perfect squares?	So the root is between	Closer to	Calculator
$\sqrt{18}$				
$\sqrt{40}$				
$\sqrt{80}$				
$\sqrt{135}$				

7. A square lawn has an area of 85 m^2.

(a) How long is each side, to the nearest tenth of a metre?

(b) Garden edging goes all the way around the lawn. How many metres of edging are needed, to the nearest tenth?

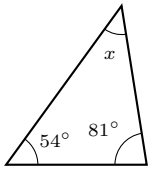
(c) The edging is sold in 6 m rolls. How many rolls must be bought?

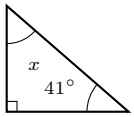
8. Find the mistake in each answer. Explain it, and give the correct answer.

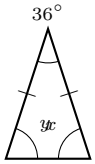
(a) Jesse wrote: $6^2 = 12$ and $\sqrt{64} = 32$.

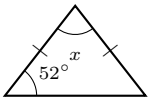
(b) Maya says a square room with an area of 18 m^2 is about 9 m long on each side.

9. Find each missing angle. Show your equation.

(a) 

(b) 

(c) 

(d) 

10. Name each triangle by its sides and by its angles.

	Side lengths	**Angles**	**By sides**	**By angles**
(a)	8 m, 8 m, 8 m	60°, 60°, 60°		
(b)	5 cm, 12 cm, 13 cm	23°, 67°, 90°		
(c)	5 cm, 5 cm, 8 cm	37°, 37°, 106°		
(d)	7.7 cm, 8.7 cm, 9.4 cm	50°, 60°, 70°		
(e)	3 cm, 5 cm, 7 cm	22°, 38°, 120°		

11. Write *always*, *sometimes* or *never*. Give a reason or an example.

(a) A triangle has two right angles. _____

(b) A right triangle is also isosceles. _____

(c) An equilateral triangle is acute. _____

(d) In a right triangle, the hypotenuse is the longest side. _____

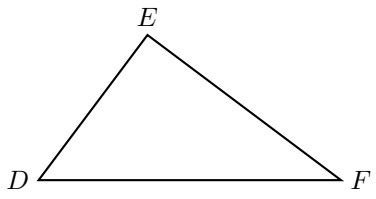
12. $\triangle DEF$ has its right angle at E.

(a) Mark the right angle. Label each side with a small letter.

(b) The hypotenuse is side _____, also called side _____.

(c) The legs are sides _____ and _____, also called _____ and _____.

(d) $DE = 9$ cm, $EF = 12$ cm and $DF = 15$ cm. How long is the hypotenuse? _____



13. Draw a sketch for each situation. Mark the right angle and label the hypotenuse.

(a) A guy wire runs from the top of a radio tower that stands straight up to a stake in the level ground. The wire makes a 58° angle with the ground. What angle does the wire make with the tower?

(b) A carpenter's speed square is a right triangle with two equal legs. What are the sizes of its other two angles?

14. $\sqrt{144}$ equals

A. 72 **B.** 12 **C.** 20 736 **D.** 288

15. A triangle has angles of 38° and 67°. The third angle is

A. 105° **B.** 75° **C.** 255° **D.** 85°

16. In right $\triangle XYZ$, the right angle is at Y. The hypotenuse is

A. side XY **B.** side YZ **C.** side XZ **D.** side x

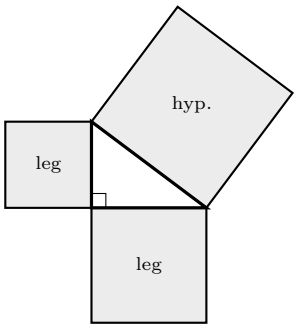
17. Numerical Response $\sqrt{95}$, rounded to the nearest tenth, is _____.

18. Numerical Response An isosceles triangle has a 28° angle between its two equal sides. Each of the other two angles is _____°.

Lesson 2: The Pythagorean Theorem: Finding the Hypotenuse

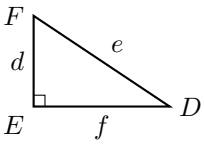
Assignment

1. A square is drawn on each side of a right triangle. Use the areas of the squares on the legs to complete the table.

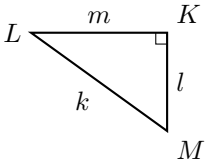


	Square on one leg	**Square on other leg**	**Square on hypotenuse**	**Length of hypotenuse**
(a)	36 cm^2	64 cm^2		
(b)	25 m^2	144 m^2		
(c)	7 in^2	9 in^2		

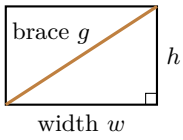
2. For each right triangle, name the hypotenuse. Then write the Pythagorean theorem using the triangle's letters.



(a) Hypotenuse: _____



(b) Hypotenuse: _____



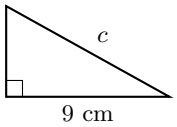
(c) A gate with a diagonal brace. Hypotenuse: _____

3. Each row is a right triangle. Complete the table. (These answers come out to whole numbers or tenths.)

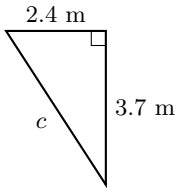
	Leg a	**Leg** b	a^2	b^2	$c^2 = a^2 + b^2$	**Hypotenuse** c
(a)	7	24				
(b)	12	16				
(c)	0.9	1.2				
(d)	11	60				

4. Find the length of the hypotenuse, to the nearest tenth. Check that each answer is reasonable.

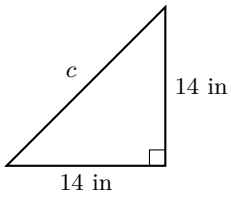
(a)



(b)



(c)



5. A right triangle has legs of 15 cm and 20 cm. Three students found the hypotenuse.

Jordan: $15 + 20 = 35$ cm **Ava:** $15^2 + 20^2 = 625$, so $c = 625$ cm **Ben:** $15^2 + 20^2 = 625$, $\sqrt{625} = 25$ cm

(a) Who is correct? _____

(b) Explain the mistake each of the other two students made.

(c) How could Jordan and Ava each tell that their answer was not reasonable?

6. Two boards, 5 ft and 12 ft long, are joined at one end by a hinge.

(a) When the hinge is open to exactly 90°, how far apart are the free ends?

(b) The hinge is closed a bit, to an angle smaller than 90°. Now the free ends are 10.4 ft apart. Work out c^2 and compare it with $5^2 + 12^2$.

(c) Use (a) and (b) to explain why the Pythagorean theorem only works for right triangles.

7. Which of these situations makes a right triangle? Write *yes* or *no*. For each *yes*, say where the right angle is.

(a) A ladder leans against the wall of a house on flat ground. _____

(b) A slice cut from a round pizza. _____

(c) A cyclist rides due west, then turns and rides due south. _____

(d) The diagonal of a rectangular phone screen. _____

8. Give one way the Pythagorean theorem was used long ago, and one way it is used today.

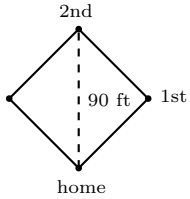
9. A wheelchair ramp rises 0.45 m over a horizontal distance of 5.4 m (a slope of 1 : 12). How long is the ramp's surface, to the nearest centimetre?

10. A guy wire runs from a point 18 m up a cell tower to a stake in the ground 7.5 m from the base of the tower.

(a) How long is the wire?

(b) The tower needs 4 wires like this. Wire costs $4.29 per metre. Find the cost of the wire before tax.

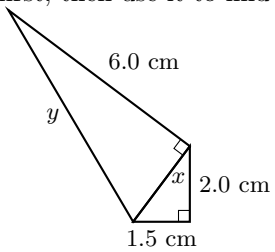
11. A baseball diamond is a square with sides of 90 ft. How far does the catcher throw from home plate to second base? Round to the nearest tenth of a foot.



12. A loading ramp for a pickup truck rises 3 ft 4 in over a horizontal distance of 10 ft. How long is the ramp, to the nearest inch? Give your answer in feet and inches.

13. Two snowmobiles leave a cabin near Grande Prairie at the same time. One travels due north at 32 km/h. The other travels due east at 45 km/h. How far apart are they after 30 minutes, in a straight line? Round to the nearest tenth of a kilometre.

14. Find x first, then use it to find y.



15. The legs of a right triangle are 10 cm and 4 cm. The hypotenuse is about

A. 6.0 cm **B.** 10.8 cm **C.** 14.0 cm **D.** 116 cm

16. In $\triangle PQR$, the right angle is at R. Which equation is correct?

A. $p^2 + r^2 = q^2$ **B.** $q^2 + r^2 = p^2$ **C.** $p^2 + q^2 = r^2$ **D.** $p + q = r$

17. Which statement is true?

A. $a^2 + b^2 = c^2$ is true for every triangle.

B. $a^2 + b^2 = c^2$ is true only for right triangles, where c is the side across from the right angle.

C. The hypotenuse is the shortest side of a right triangle.

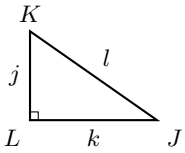
D. In a right triangle, the hypotenuse is equal to the two legs added together.

18. Numerical Response A laptop screen is 13.6 in wide and 8.5 in high. The length of its diagonal, to the nearest tenth of an inch, is _____ in.

Lesson 3: The Pythagorean Theorem: Finding a Leg, and Squaring a Corner

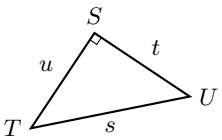
Assignment

1. For each right triangle, name the hypotenuse and write the Pythagorean theorem using its letters.

(a) 

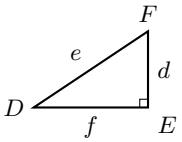
Hypotenuse: _____

_____

(b) 

Hypotenuse: _____

_____

(c) 

Hypotenuse: _____

_____

2. Find the missing leg. Each triangle is a right triangle.

(a) hypotenuse 25 cm, one leg 7 cm

(b) hypotenuse 61 m, one leg 11 m

(c) hypotenuse 34 in, one leg 16 in

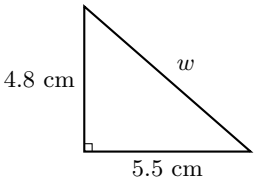
3. Find the missing leg, to the nearest tenth.

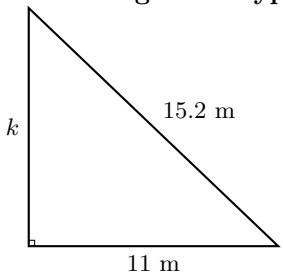
(a) hypotenuse 14.5 cm, one leg 8.2 cm

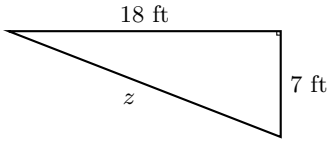
(b) hypotenuse 9.6 m, one leg 3.9 m

(c) hypotenuse 20 in, one leg 16.4 in

4. For each triangle, write whether the missing side is a **leg** or the **hypotenuse**. Then find it to the nearest tenth.

(a) 

(b) 

(c) 

5. A right triangle has a hypotenuse of 10 cm and one leg of 6 cm. Tyler found the other leg like this:

$$6^2 + 10^2 = x^2 \qquad x^2 = 136 \qquad x \approx 11.7 \text{ cm}$$

(a) How can you tell, without doing any math, that Tyler's answer is wrong?

(b) What was his mistake? Find the correct length.

6. The size of a TV is the length of its diagonal d. A TV screen has width w and height h.

(a) Write the Pythagorean theorem for the screen using d, w and h.

(b) Rearrange your formula to find h.

(c) A 55 in TV is 47.9 in wide. Use your formula to find its height, to the nearest tenth of an inch.

7. A painter leans a 5.5 m ladder against a wall. The foot of the ladder is 1.4 m from the wall.

(a) Draw a sketch. How high up the wall does the ladder reach, to the nearest tenth of a metre?

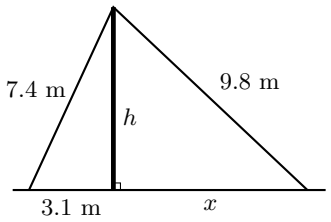
(b) Is the ladder set up safely by the 4-to-1 rule? Explain.

8. Kai flies a drone over a field near Strathmore. His controller shows that the drone is 120 m from him in a straight line. The drone is directly above a fence post that is 75 m from Kai. (Ignore Kai's height.)

(a) How high is the drone, to the nearest tenth of a metre?

(b) In Canada, most drones must stay below 122 m. Is Kai's drone below that height?

9. A tall pole stands straight up in the middle of a big event tent. Two ropes run from the top of the pole to stakes in the ground. The left rope is 7.4 m long and is staked 3.1 m from the bottom of the pole. The right rope is 9.8 m long.



(a) How tall is the pole, to the nearest tenth of a metre?

(b) How far from the pole is the right rope staked? Keep the full value from (a).

10. Is each triangle a right triangle? Show your work.

(a) 20 cm, 48 cm, 52 cm

(b) 5 m, 11 m, 12 m

(c) 2.4 ft, 1.0 ft, 2.6 ft

(d) 10 in, 10 in, 14 in

11. Ana tested a triangle with sides 12 cm, 13 cm and 5 cm. She wrote: "$12^2 + 13^2 = 313$ and $5^2 = 25$. They are not equal, so it is not a right triangle." Is Ana right? Find her mistake and correct it.

12. Wei is laying out the floor frame for a garden shed.

(a) Complete the table of 3-4-5 multiples.

Multiply by	5	7	10	0.3
Sides				

(b) Wei measures 3 ft along one side and 4 ft along the other. The diagonal between the marks is 4 ft 10 in. Is the corner square? What should Wei do?

(c) Wei's boards are 12 ft long. Which multiple of 3-4-5 would give a more accurate check than 3-4-5? Why?

13. A grounds crew paints the lines of a soccer field in Lethbridge. The field is a rectangle 100 m long and 64 m wide.

(a) How long should each diagonal be, to the nearest tenth of a metre?

(b) The crew measures the two diagonals: 118.7 m and 119.4 m. What does this tell them?

14. Which set of side lengths makes a right triangle?

A. 4, 5, 6 **B.** 5, 12, 14 **C.** 7, 24, 25 **D.** 9, 10, 14

15. A 17 ft ladder leans against a wall with its foot 8 ft from the wall. How high up the wall does it reach?

A. 9 ft **B.** 15 ft **C.** 18.8 ft **D.** 25 ft

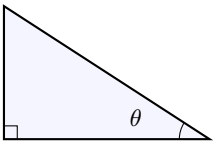
16. Numerical Response A farm gate is a rectangle 1.2 m high. A diagonal brace runs from one corner to the opposite corner and is 2.15 m long. The width of the gate, to the nearest hundredth of a metre, is _____ m.

17. Numerical Response A deck frame is a rectangle 18 ft by 24 ft. To check that it is square, a builder measures a diagonal. If the corners are square, the diagonal is _____ ft.

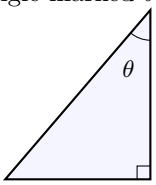
Lesson 4: Similar Right Triangles and the Tangent Ratio

Assignment

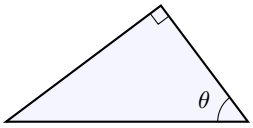
1. Write **opp**, **adj** or **hyp** beside each side. Use the angle marked θ.



(a)



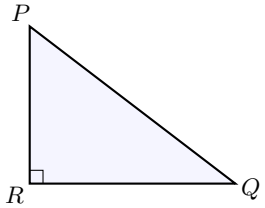
(b)



(c)

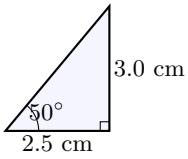
2. In $\triangle PQR$, $\angle R = 90°$. Name each side with two letters.

	opp	**adj**	**hyp**
(a) from $\angle P$			
(b) from $\angle Q$			

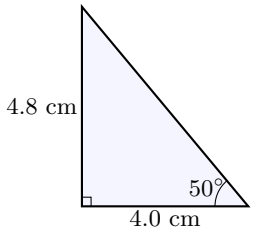


(c) Write $\tan P$ and $\tan Q$ as ratios of sides.

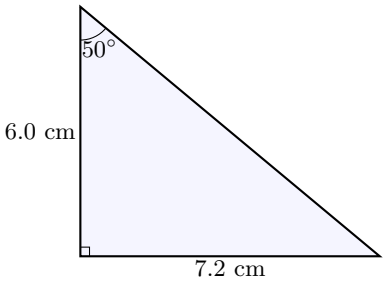
3. These three right triangles each have a 50° angle. (They are not drawn to scale.)



Triangle 1



Triangle 2



Triangle 3

(a) Why are the triangles similar?

(b) Use the 50° angle to complete the table. Careful: in Triangle 3, which side is opp?

	Triangle 1	**Triangle 2**	**Triangle 3**
opp (cm)			
adj (cm)			
opp ÷ adj			

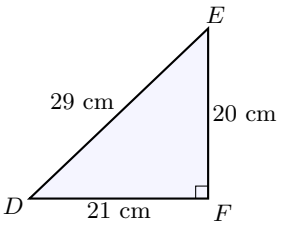
(c) Find $\tan 50°$ with your calculator (4 decimal places). Does it agree with the table?

(d) Another right triangle with a 50° angle has adj = 10 m. Use the table to predict opp.

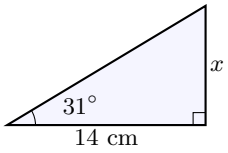
4. (a) Use your calculator in DEG mode. Round to 4 decimal places.

θ	15°	41°	75°	89°
$\tan\theta$				

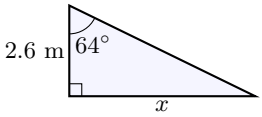
(b) In $\triangle DEF$, $\angle F = 90°$, $DF = 21$ cm, $EF = 20$ cm and $DE = 29$ cm. Write $\tan D$ and $\tan E$ as fractions and as decimals (4 decimal places).



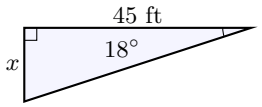
5. Find x, to the nearest tenth.



(a)

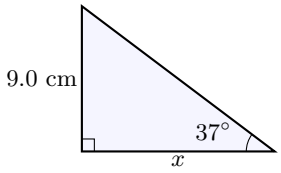


(b)

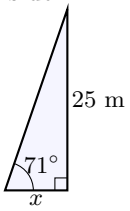


(c)

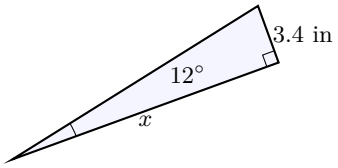
6. Find x, to the nearest tenth. Here x is the **adjacent** side.



(a)

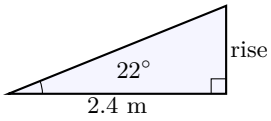


(b)

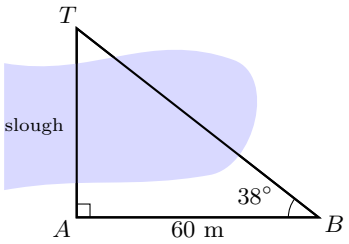


(c)

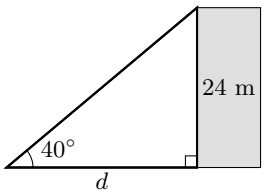
7. The roof of a garden shed slopes up at 22° from the horizontal. It covers a horizontal distance (the run) of 2.4 m. How much does the roof rise? Round to the nearest hundredth of a metre.



8. Farmer Dana wants to know how wide a slough is, from point A to a fence post T on the other side. She walks 60 m along the shore from A to B, at a right angle to AT. From B, the angle between BA and the line to the post is 38°. How wide is the slough? Round to the nearest tenth.



9. A photographer wants a photo of a 24 m tall office building. She wants the line from her camera on the ground to the top of the building to make a 40° angle with the ground. How far from the building should she put the camera? Round to the nearest tenth.

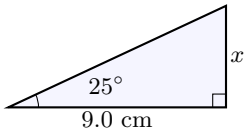


10. A ramp at a loading dock must rise 0.35 m. Its angle with the ground will be 4.5°.

(a) How far along the ground does the ramp reach? Round to the nearest tenth of a metre.

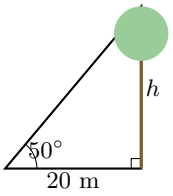
(b) A ramp with a 1 : 12 slope reaches 12 times its rise along the ground. Is this ramp gentler or steeper than 1 : 12? Explain.

11. Zoe wants the opposite side x in this triangle. She writes:
$x = 9.0 \div \tan 25° = 19.3\ cm$
(a) Without a calculator, explain why her answer can't be right.



(b) Find the correct value of x.

12. Standing on the ground 20 m from a tree, Malik measures a 50° angle from the ground to the top of the tree.
(a) How tall is the tree? Keep the full calculator value.

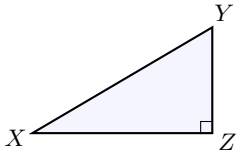


(b) Malik wants the angle to be only 35°. How far from the tree must he stand? How much farther back is that? Round to the nearest tenth.

13. A roof with a pitch of 5/12 rises 5 units for every 12 units it runs across. Is a 5/12 roof steeper or less steep than a roof that slopes at 24°? Explain using tangents.

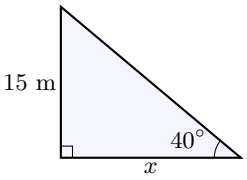
14. In $\triangle XYZ$, $\angle Z = 90°$. Which ratio is $\tan X$?

A. $\dfrac{YZ}{XZ}$ **B.** $\dfrac{YZ}{XY}$ **C.** $\dfrac{XZ}{XY}$ **D.** $\dfrac{XZ}{YZ}$



15. Which equation gives x?

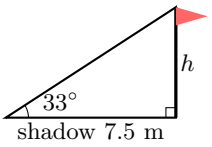
A. $x = 15 \times \tan 40°$ **B.** $x = \dfrac{15}{\tan 40°}$ **C.** $x = \dfrac{\tan 40°}{15}$ **D.** $x = \dfrac{15}{40}$



16. A right triangle has a 62° angle. The side adjacent to it is 5.0 m. The side opposite the 62° angle is about

A. 2.7 m **B.** 5.0 m **C.** 9.4 m **D.** 10.7 m

17. Numerical Response A flagpole stands straight up on level ground. Its shadow is 7.5 m long when the sun's rays make a 33° angle with the ground. The height of the flagpole, to the nearest tenth of a metre, is _____ m.



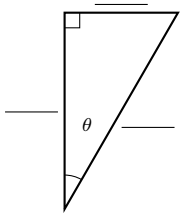
18. Numerical Response A kite is flying 18 m above the ground, straight above point P. From where Noah stands on the ground, the angle up to the kite is 56°. Noah's distance from P, to the nearest tenth of a metre, is _____ m.

Lesson 5: The Sine Ratio

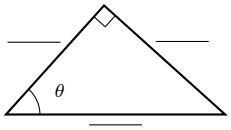
Assignment

Use your calculator in DEG mode. Round lengths to the nearest tenth and sine values to 4 decimal places unless a question says otherwise.

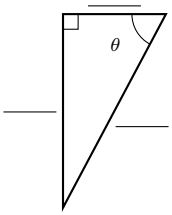
1. Label each side opp, adj or hyp for the angle θ.



(a)

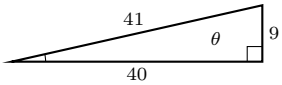


(b)

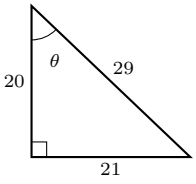


(c)

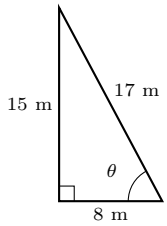
2. Write $\sin\theta$ as a fraction, then as a decimal.



(a)



(b)



(c)

3. Use your calculator. Round to 4 decimal places.

(a) $\sin 12° =$ _____ (b) $\sin 38° =$ _____ (c) $\sin 55° =$ _____ (d) $\sin 81° =$ _____

(e) Which of these four angles has the sine closest to 1? Why does that make sense?

4. A skatepark has three ramps of different lengths. All three make a 25° angle with the ground.

Ramp	**Length (hyp)**	**Height (opp)**	**opp ÷ hyp** (2 decimals)
A	5.0 m	2.1 m	
B	9.0 m	3.8 m	
C	11.0 m	4.6 m	

(a) Fill in the last column.

(b) What do you notice? Find $\sin 25°$ and compare.

(c) A fourth ramp is 20 m long, also at 25°. Predict its height.

5. True or false? If false, explain why.

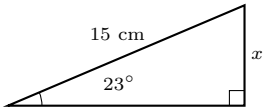
(a) For every acute angle, $\sin\theta$ is less than 1. _____

(b) A bigger right triangle with a 50° angle has a bigger value of $\sin 50°$. _____

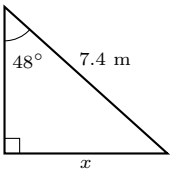
(c) $\sin 60°$ is double $\sin 30°$. _____

(d) If $\sin\theta = 0.5$, the opposite side is half as long as the hypotenuse. _____

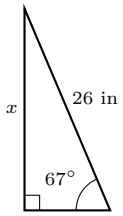
6. Find the length of the side x (opposite the angle).



(a)

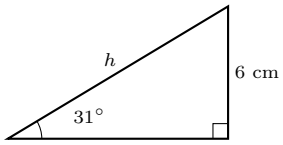


(b)

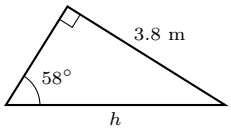


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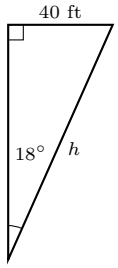
7. Find the length of the hypotenuse h.



(a)

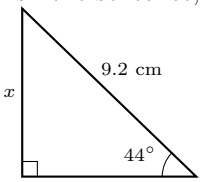


(b)

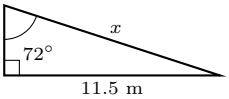


(c) (not to scale)

8. For each triangle, finish the sentence, then solve for x.



(a) The unknown is on the _____, so I _____.



(b) The unknown is on the _____, so I _____.

9. A farmer sets a 16 m grain auger against a bin. The auger makes a 33° angle with the ground, and its top end rests at the top of the bin. Sketch a right triangle. How tall is the bin?

10. A conveyor belt at a gravel pit lifts gravel to the top of a pile 6.5 m high. The belt makes a 24° angle with the ground. Sketch a right triangle. How long is the belt, from the ground to the top of the pile?

11. A parasailer is towed behind a boat on a 75 m rope. The rope makes a 38° angle with the horizontal. It is tied to the boat 1.5 m above the water. How high above the water is the parasailer? (Treat the rope as straight.)

12. Checking answers.

(a) A right triangle has a hypotenuse of 13 cm and a 42° angle. Dev wants the side opposite the 42° angle. His work: $x = 13 \div \sin 42° \approx 19.4$ cm. How can you tell, without a calculator, that his answer is wrong?

(b) What did Dev do wrong? Find the correct answer.

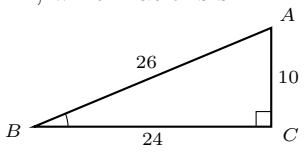
(c) Another right triangle has a hypotenuse of 20 m and a 28° angle. Three students found the side opposite the 28° angle: 9.4 m, 17.7 m, 42.6 m. Use $\sin 30° = 0.5$ to decide which answer is reasonable, without finding the exact answer. Explain.

13. A zipline starts on a platform 25 m above the ground and ends on a platform 3.5 m above the ground. The cable makes a 6° angle with the horizontal. Sketch a diagram. How long is the cable? (Treat it as straight.)

14. Nadia leans a 5.5 m ladder against a wall at 68° to the ground. Her neighbour tells her to move the foot closer to the wall so the angle is 75°. How much higher up the wall does the ladder reach after she moves it? Round to the nearest

hundredth of a metre.

15. For $\angle B$, which ratio is $\sin B$?



A. $\frac{10}{26}$ **B.** $\frac{24}{26}$ **C.** $\frac{10}{24}$ **D.** $\frac{26}{10}$

16. A right triangle has an opposite side of 7 cm and a 36° angle. Which equation gives the hypotenuse h?

A. $h = 7 \times \sin 36°$ **B.** $h = \frac{7}{\sin 36°}$ **C.** $h = \frac{\sin 36°}{7}$ **D.** $h = 7 + \sin 36°$

17. Jaden presses [sin] 30 [)] [ENTER] and his calculator shows -0.9880. What is the most likely reason?

A. He should have pressed [2nd] first.

B. His calculator is in RAD mode, not DEG mode.

C. He forgot to type the brackets before 30.

D. The sine of 30° is a negative number.

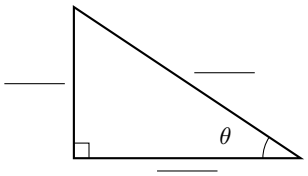
18. Numerical Response A boat-launch ramp rises 1.8 m from the water to the parking lot. The ramp makes a 9° angle with the horizontal. The length of the ramp surface, to the nearest tenth of a metre, is _____ m.

Lesson 6: The Cosine Ratio

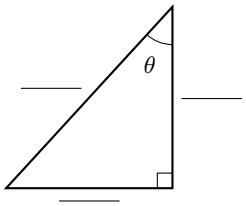
Assignment

1. Label the sides of each triangle **hyp**, **opp** and **adj** from the angle θ.

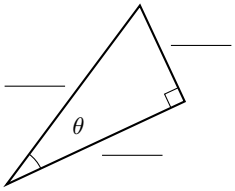
(a)



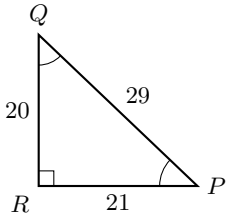
(b)



(c)



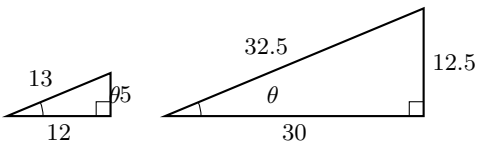
2. In $\triangle PQR$, $\angle R = 90°$.



(a) Write $\cos P$ as a fraction and as a decimal to 4 places.

(b) Write $\cos Q$ as a fraction and as a decimal to 4 places.

3. The two right triangles below both have the angle θ.



(a) Find $\dfrac{\text{adj}}{\text{hyp}}$ for each triangle, to 4 decimal places.

(b) Why are the two answers the same?

(c) Find $\cos 22°$ and $\cos 23°$. Which is closer to the size of θ?

4. Find each value to 4 decimal places.

(a) $\cos 12° =$ _____ (b) $\cos 48° =$ _____ (c) $\cos 66° =$ _____ (d) $\cos 81° =$ _____

(e) Without a calculator, which is bigger, $\cos 25°$ or $\cos 52°$? Explain.

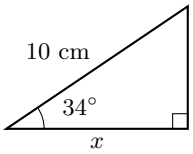
5. Riley presses [cos] 60 [)] [ENTER] and gets -0.952413.

(a) How do you know this is wrong without doing any math?

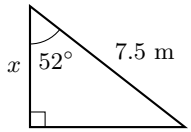
(b) What should Riley fix on the calculator?

6. Find x. Round to the nearest tenth.

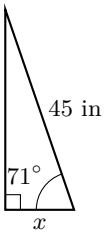
(a)



(b)

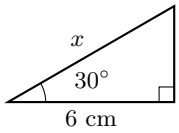


(c)

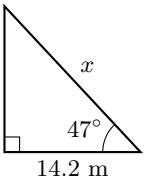


7. Find x. Round to the nearest tenth.

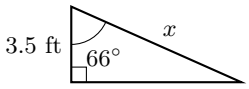
(a)



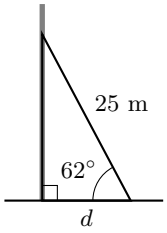
(b)



(c)

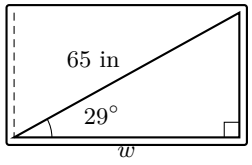


8. A guy wire runs from the top of a radio tower to an anchor in the ground. The wire is 25 m long and makes a 62° angle with the ground. How far is the anchor from the base of the tower? Round to the nearest tenth of a metre.



9. Mila flies a kite on a straight string. The string makes a 48° angle with the ground. The kite is right above a fence post that is 35 m from Mila (measured along the ground). Sketch the triangle, then find the length of the string to the nearest tenth of a metre. (Treat Mila's hand as being at ground level.)

10. A TV is sold as a 65-inch TV: its diagonal is 65 in. The diagonal makes a 29° angle with the bottom edge of the screen.



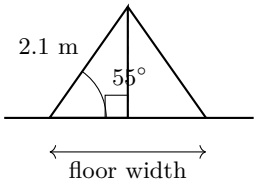
(a) How wide is the screen, to the nearest tenth of an inch?

(b) Will the TV fit in a cabinet opening that is 58 in wide? (The frame adds about half an inch on each side.)

11. A chairlift cable runs in a straight line up a ski hill at 22° to the horizontal. On a map, the horizontal distance from the bottom station to the top station is 1150 m. Sketch the triangle, then find the length of the cable to the nearest metre.

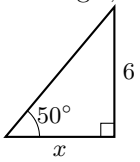
12. The front of an A-frame tent is a triangle. Each sloping side is 2.1 m long and makes a 55° angle with the ground.

(a) How wide is the tent floor, to the nearest tenth of a metre?

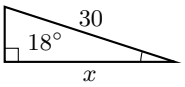


(b) Can two sleeping pads that are each 0.65 m wide lie side by side on the floor?

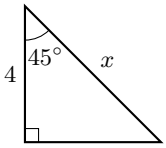
13. For each triangle, name the side you know and the side you want (x). Then choose sin, cos or tan. Do not solve.



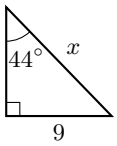
(a) Know: _____
Want: _____
Use: _____



(b) Know: _____
Want: _____
Use: _____

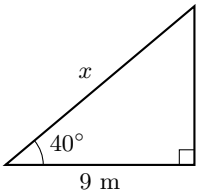


(c) Know: _____
Want: _____
Use: _____



(d) Know: _____
Want: _____
Use: _____

14. Kai has to find x in this triangle. Here is his work:



$$\cos 40° = \frac{9}{x}$$
$$x = 9 \times \cos 40°$$
$$x \approx 6.9 \text{ m}$$

(a) How can you tell that his answer is wrong without redoing the question?

(b) What was his mistake? Find the correct value of x.

15. Canadian accessibility guides suggest that a wheelchair ramp should slope at about 4.8° or less (a slope of about 1 : 12). A ramp at 4.8° has a horizontal run of 9.0 m.

(a) How long is the ramp surface, to the nearest tenth of a metre?

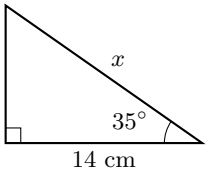
(b) How high does the ramp rise? Choose the right ratio. Round to the nearest hundredth of a metre.

(c) Check (b): a 1 : 12 slope rises 1 m for every 12 m of run. Is your answer reasonable?

16. In $\triangle ABC$, $\angle C = 90°$, $a = 12$, $b = 35$ and $c = 37$. What is $\cos B$?

A. $\frac{35}{37}$ **B.** $\frac{12}{37}$ **C.** $\frac{12}{35}$ **D.** $\frac{35}{12}$

17. Which expression gives x?



A. $14 \times \cos 35°$

B. $14 \div \cos 35°$

C. $\cos 35° \div 14$

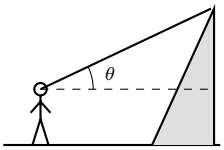
D. $14 \times \sin 35°$

18. Numerical Response A moving truck's loading ramp is 3.2 m long and makes an 18° angle with the ground. The bottom of the ramp touches the ground _____ m from the back of the truck, to the nearest tenth of a metre.

Lesson 7: Angles of Elevation and Depression

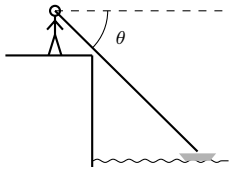
Assignment

1. Is the marked angle θ an angle of elevation, an angle of depression, or neither?



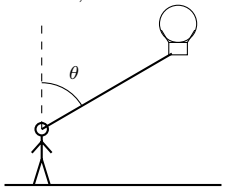
(a) top of a ski jump

_____



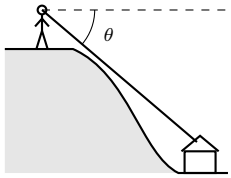
(b) from a bridge

_____



(c) a hot-air balloon

_____



(d) a house from a hill

_____

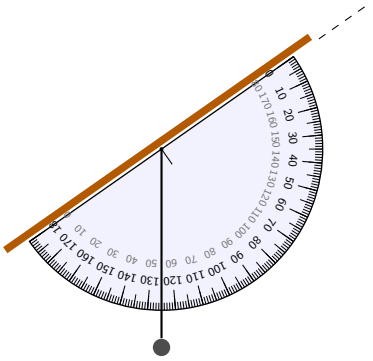
2. Fill in each blank.

(a) From a bridge, the angle of depression to a boat is 19°. The angle of elevation from the boat to the bridge is _____.

(b) From the bottom of a ski hill, the angle of elevation to the chalet at the top is 27°. The angle of depression from the chalet to the bottom is _____.

(c) Ella looks down from a balcony. Her line of sight makes an angle of 58° with the wall of the building (the vertical). The angle of depression is _____, because _____.

3. (a) Read the clinometer and find the angle of elevation.



Reading: _____

Angle of elevation: _____

(b) Find the angle of elevation for each reading.

Reading	70°	55°	104°	90°	128°
Angle					

4. For each situation: sketch it, label the right triangle (opp, adj, hyp from the angle), write the ratio you need, then solve. Round to the nearest tenth.

(a) From a point 30 m from the base of a cell tower, the angle of elevation to the top is 56°. How tall is the tower? (Measure from the ground.)

(b) From a hot-air balloon, the angle of depression to a landing field is 28°. The distance from the balloon to the field along the line of sight is 400 m. How high is the balloon?

5. Hana stands 15 m from the base of a spruce tree. Her eyes are 1.5 m above the ground. The angle of elevation to the top of the tree is 52°. How tall is the tree, to the nearest tenth of a metre?

6. A zipline cable is 85 m long. From the top tower, the cable goes down at an angle of depression of 12°. Assume the cable is straight. Round to the nearest tenth.

(a) How much lower is the bottom end of the cable than the top end?

(b) What is the horizontal distance between the two ends?

7. A ranger in a fire lookout tower in the foothills is 22 m above the ground. She sees a deer at an angle of depression of 8°. How far is the deer from the base of the tower, to the nearest metre?

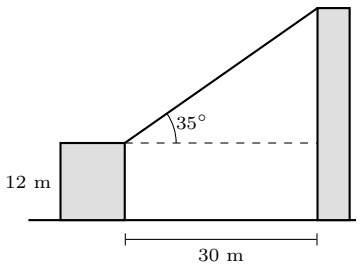
8. Priya flies a kite on 75 m of string. She holds the string 1.1 m above the ground, and the string makes an angle of elevation of 40°. Assume the string is straight. How high is the kite above the ground, to the nearest tenth of a metre?

9. A small plane is 950 m above the runway at Red Deer. It comes in to land in a straight line at an angle of depression of 6°.

(a) How far does the plane fly along its path to touch down? Round to the nearest tenth of a kilometre.

(b) Your answer is much longer than 950 m. Why does that make sense?

10. Theo is on the roof of a 12 m apartment building. An office tower is 30 m away. From the roof, the angle of elevation to the top of the tower is 35°. How tall is the office tower, to the nearest tenth of a metre? (Measure from the roof; ignore Theo's height.)



11. Mateo stands 20 m from a building. His eyes are 1.7 m above the ground. The angle of elevation to the roof is 35°. Here is his work:

$h = 20 \times \tan 35° = 14.0$ m. The building is 14.0 m tall.

(a) What did Mateo forget?

(b) Find the correct height of the building.

12. Lena stands 40 m from a grain elevator. The angle of elevation to the top is 60°. She says the top is 23.1 m above her eyes.

(a) Without a calculator, explain why her answer cannot be right.

(b) Find the correct height above her eyes, to the nearest tenth.

13. Jaxon stands 18 m from the wall of the school gym. He looks through his clinometer at the top of the wall, and the string crosses the scale at 61°. His eyes are 1.5 m above the ground. How tall is the wall, to the nearest tenth of a metre?

14. From a sailboat, the angle of elevation to the top of a lighthouse is 18°. What is the angle of depression from the top of the lighthouse to the sailboat?

A. 72° **B.** 18° **C.** 162° **D.** 108°

15. From the top of a 60 m cliff, the angle of depression to a boat is 23°. Which expression gives the distance from the boat to the base of the cliff?

A. $60 \times \tan 23°$ **B.** $\dfrac{60}{\tan 23°}$ **C.** $60 \times \sin 23°$ **D.** $\dfrac{60}{\sin 23°}$

16. Numerical Response A guy wire is attached to a radio tower 40 m above the ground. From its anchor on the ground, the angle of elevation of the wire is 63°. The length of the wire, to the nearest tenth of a metre, is _____ m.

17. Numerical Response A wheelchair ramp is 7.2 m long. It rises at an angle of elevation of about 4.8°. How high does the ramp rise, to the nearest tenth of a **centimetre**? _____ cm

Lesson 8: Finding Angles with Inverse Trig Ratios

Assignment

1. Find each angle θ to the nearest degree.

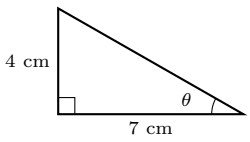
(a) $\sin\theta = 0.42$ (b) $\cos\theta = 0.81$ (c) $\tan\theta = 0.84$

(d) $\sin\theta = 0.974$ (e) $\cos\theta = 0.12$ (f) $\tan\theta = 5.67$

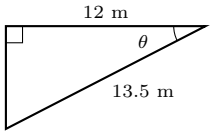
2. Find each angle to the nearest degree. Type the whole fraction.

(a) $\sin A = \frac{2}{7}$ (b) $\cos B = \frac{11}{15}$ (c) $\tan C = \frac{9}{4}$ (d) $\tan D = \frac{3}{11}$

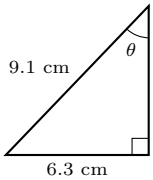
3. For each triangle, label the two given sides opp, adj or hyp from θ, name the ratio you will use, and find θ to the nearest degree.



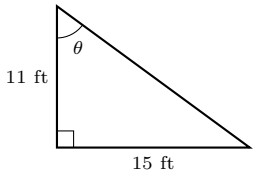
(a)



(b)



(c)



(d)

4. One acute angle of a right triangle is given. Find the other acute angle.

(a) 34° _____ (b) 71° _____ (c) 12.5° _____

5. In $\triangle ABC$, $\angle C = 90°$, $a = 7.5$ cm and the hypotenuse $c = 10$ cm.

(a) Find $\angle A$. (b) Find $\angle B$ by subtracting. (c) Check $\angle B$ with a ratio.

6. A right triangle has sides 7 cm, 24 cm and 25 cm. Let θ be the angle across from the 7 cm side.

(a) Find θ with the tangent ratio.

(b) Find θ again with the sine ratio. Do you get the same angle?

(c) Find the third angle of the triangle.

7. Without a calculator, decide whether θ is less than, equal to, or more than 45°. Explain. Then check (a) and (b) with a calculator.

(a) $\tan\theta = \frac{7}{5}$ (b) opp = 3.8 m and adj = 4.1 m

(c) Both legs are 6.5 cm. (d) opp = 4 cm and hyp = 8 cm

8. A right triangle has opp = 9 m and hyp = 14 m from angle θ.

(a) Liam pressed [2nd] [sin] 14 [÷] 9 [)] [ENTER] and got ERROR. What did he do wrong?

(b) Priya typed the correct keys but got 0.6982. What is wrong?

(c) Find θ correctly.

9. Jayden leans a 7.3 m ladder against a house. The foot of the ladder is 2.1 m from the wall.

(a) What angle does the ladder make with the ground?

(b) Is this close to the safe angle of about 75°? Should he move the foot of the ladder closer to the wall or farther away?

(c) **Challenge:** How far from the wall should the foot be for an angle of exactly 75°? Round to the nearest tenth of a metre.

10. Two wheelchair ramps are being checked. Find each angle to the nearest tenth of a degree. Does it meet the guideline of about 4.8° or less?

(a) Ramp A rises 0.45 m over a horizontal distance of 5.8 m.

(b) Ramp B rises 0.5 m, and the ramp surface is 5.2 m long.

11. A roofer uses pitch to describe how steep a roof is.

(a) Complete the table. Round to the nearest degree.

Pitch	3/12	6/12	9/12	12/12
Angle with the horizontal				

(b) Erin says, "A 12/12 roof is 4 times as steep as a 3/12 roof, so its angle is 4 times as big." Is she right? Use the table.

12. Find each angle of depression to the nearest degree. Draw a sketch first.

(a) A zipline cable is 120 m long. The bottom end is 18 m lower than the top end. Find the angle of depression of the cable at the top.

(b) A lighthouse keeper's window is 42 m above the water. She sees a boat that is 350 m from the base of the lighthouse. Find the angle of depression to the boat.

13. A guy wire is attached to a radio tower 30 m above the ground. It is anchored to the ground 14 m from the base of the tower.

(a) Find the angle between the wire and the ground.

(b) Find the angle between the wire and the tower.

14. A ski run at a hill near Calgary is 1.2 km long, measured along the slope. From top to bottom, it drops 380 m. Find the angle of the slope to the nearest degree. **Hint:** use the same units for both lengths.

15. In a right triangle, the leg opposite θ is 3 cm and the leg adjacent to θ is 8 cm. To the nearest degree, θ is

A. 21° **B.** 69° **C.** 22° **D.** 68°

16. Kai tries to find an angle and types [2nd] [cos] 1.4 [)] [ENTER]. The calculator shows ERROR. The most likely mistake is that he

A. has his calculator in RAD mode

B. divided the hypotenuse by the adjacent side instead of the adjacent side by the hypotenuse

C. should have pressed [2nd] [tan] instead

D. is looking for an angle bigger than 90°

17. Numerical Response A ramp into a moving truck is 3.2 m long and rises 0.25 m. The angle between the ramp and the ground, to the nearest tenth of a degree, is _____°.

18. Numerical Response Zoe flies a kite on a 55 m string. The kite is directly above a point on the ground 31 m away from her. The angle between the string and the ground, to the nearest degree, is _____°.

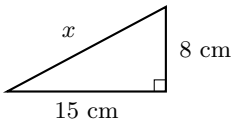
Lesson 9: Solving Right Triangles: Choosing the Right Ratio

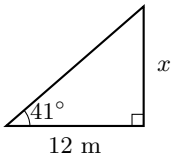
Assignment

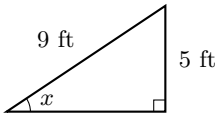
1. Which tool would you use? Write *Pythagoras*, *sine*, *cosine*, *tangent*, *inverse sine*, *inverse cosine*, *inverse tangent* or $90° -$ *angle*.

	You know	**You want**	**Tool**
(a)	the two legs	the hypotenuse	
(b)	an acute angle and the hypotenuse	the side opposite the angle	
(c)	an acute angle and the side adjacent to it	the side opposite the angle	
(d)	the side opposite θ and the hypotenuse	the angle θ	
(e)	one acute angle	the other acute angle	
(f)	one leg and the hypotenuse	the other leg	
(g)	an acute angle and the hypotenuse	the side adjacent to the angle	

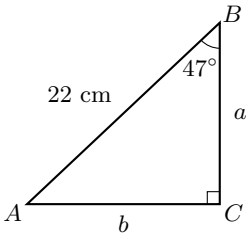
2. Name the tool and write the equation. Then find x. Round sides to the nearest tenth and angles to the nearest degree.

(a) 

(b) 

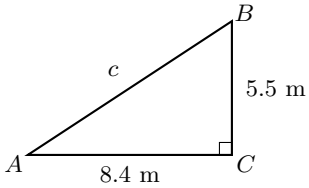
(c) 

3. Solve $\triangle ABC$: find $\angle A$, a and b.



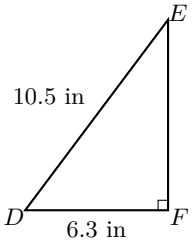
$\angle A =$ _____ $a =$ _____ $b =$ _____

4. Solve $\triangle ABC$: find c, $\angle A$ and $\angle B$.



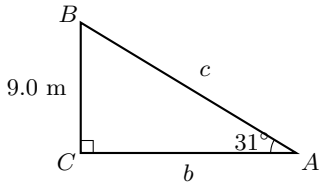
$c =$ _____ $\angle A =$ _____ $\angle B =$ _____

5. Solve $\triangle DEF$, where $\angle F = 90°$: find EF, $\angle D$ and $\angle E$.



$EF =$ _____ $\angle D =$ _____ $\angle E =$ _____

6. Solve $\triangle ABC$: find $\angle B$, b and c. (Watch where the unknown goes!)



$\angle B =$ _____ $b =$ _____ $c =$ _____

7. A 6.5 m ladder leans against a wall. It makes an angle of 74° with the ground.

(a) Sketch and label the triangle.

(b) How high up the wall does the ladder reach?

(c) How far is the foot of the ladder from the wall?

8. A zipline at a summer camp in Kananaskis starts on a platform 18 m higher than where it ends. The horizontal distance between the two ends is 150 m.

(a) Find the angle the cable makes with the horizontal, to the nearest degree.

(b) Find the length of the cable (assume it is straight).

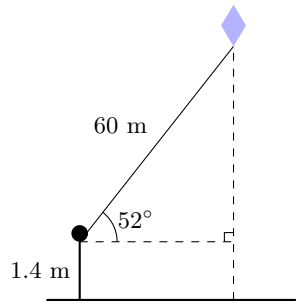
(c) Is your answer to (b) reasonable? Explain.

9. Tomas is flying a kite on a 60 m string. The string makes an angle of elevation of 52°. He holds the string 1.4 m above the ground. Assume the string is straight.

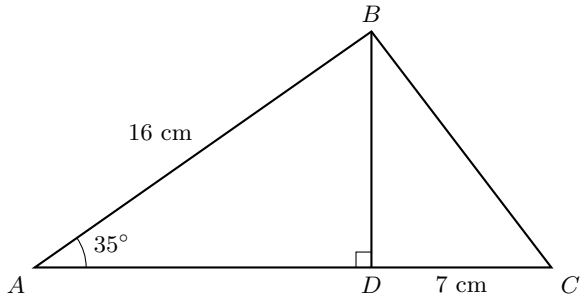
(a) How high is the kite above Tomas's hand?

(b) How high is the kite above the ground?

(c) How far is Tomas from the spot on the ground right under the kite?



10. The two right triangles share side BD. $\angle A = 35°$, $AB = 16$ cm and $DC = 7$ cm.



(a) Which triangle do you solve first? Why?

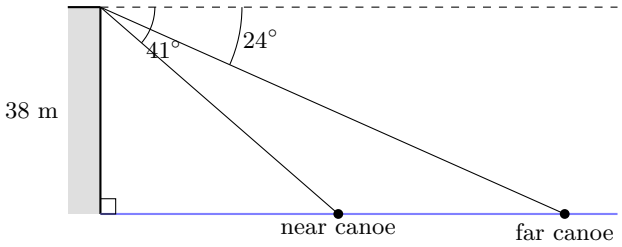
(b) Find BD.

(c) Find AD.

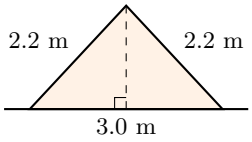
(d) Find BC.

(e) Find $\angle C$.

11. A lookout stands on top of a cliff 38 m above the water at Waterton Lake. Two canoes are lined up straight out from the bottom of the cliff. The angle of depression to the nearer canoe is 41° and to the farther canoe is 24°. How far apart are the canoes? (Hint: the angle of depression from the top equals the angle of elevation from the canoe.)



12. The front of a camping tent is an isosceles triangle. Each sloped side is 2.2 m long and the base is 3.0 m wide. A pole goes straight up the middle.



(a) How tall is the centre pole?

(b) What angle does each side make with the ground?

(c) What is the angle at the top of the tent?

13. Mila is solving a right triangle with $\angle A = 40°$ and a hypotenuse of 12 m. To find the leg *opposite* $\angle A$, she wrote $\cos 40° = \dfrac{x}{12}$ and got $x = 9.2$ m.

(a) Explain why her answer can't be right, without using a calculator.

(b) Find her mistake and the correct answer.

14. Without solving, decide whether each statement is reasonable. Explain.

(a) The legs of a right triangle are 6 m and 8 m, and Reid says the hypotenuse is 7.5 m.

(b) Tess found that $\cos\theta = 1.25$ for an acute angle θ.

(c) In a right triangle, Omar says the two acute angles are 15° and 65°.

(d) A ramp 4 m long rises at 30°. Kenji says it rises 2 m.

15. In a right triangle, the hypotenuse is 15 cm and one acute angle is 28°. Which equation gives x, the leg adjacent to the 28° angle?

A. $x = 15\sin 28°$ **B.** $x = 15\cos 28°$ **C.** $x = 15 \div \cos 28°$ **D.** $x = 15\tan 28°$

16. A loading ramp is 3.0 m long and rises 0.9 m. To the nearest tenth of a degree, the angle between the ramp and the ground is

A. 17.5° **B.** 16.7° **C.** 72.5° **D.** 0.3°

17. Numerical Response In $\triangle ABC$, $\angle C = 90°$, $a = 11$ m and $b = 6$ m. The measure of $\angle B$, to the nearest degree, is _____°.

18. Numerical Response A guy wire runs from a point 24 m up a radio tower to an anchor in the ground. The wire makes an angle of 63° with the ground. The length of the wire, to the nearest tenth of a metre, is _____ m.

Lesson 10: Unit 7 Review

Assignment

This review is set up like the unit test. Show your work on the written questions. Use your calculator in DEG mode. Round lengths to the nearest tenth and angles to the nearest degree unless a question says otherwise.

Part A: Written Response

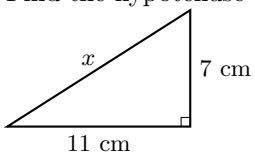
1. Fill in the blanks.

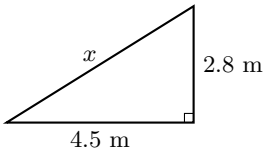
(a) $13^2 =$ _____ (b) $\sqrt{196} =$ _____ (c) $2.5^2 =$ _____

(d) $\sqrt{60}$ is between _____ and _____. To the nearest tenth, $\sqrt{60} \approx$ _____.

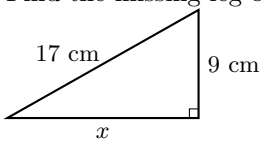
(e) A triangle has angles of 112° and 35°. The third angle is _____, and the triangle is _____.

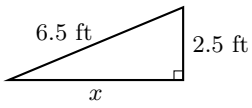
2. Find the hypotenuse of each right triangle.





3. Find the missing leg of each right triangle.





4. Is each triangle a right triangle? Show your work and explain.

(a) sides 28 m, 45 m and 53 m

(b) sides 8 in, 11 in and 14 in

5. Phone screens are sold by the length of their diagonal, in inches. A phone screen is 6.9 cm wide and 15.0 cm tall.

(a) Find the diagonal of the screen in centimetres, to the nearest hundredth.

(b) Change the diagonal to inches (1 in = 2.54 cm). What size would the ad say?

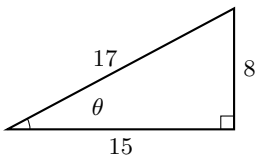
6. Marcus is laying out a backyard rink 30 ft wide and 40 ft long with string lines.

(a) What should each diagonal measure if the corners are square?

(b) He measures one diagonal: 49 ft 2 in. Is that corner square? Is the angle too big or too small?

(c) Using the 3-4-5 method, what diagonal should he get if he measures 9 ft along one side and 12 ft along the other?

7. For angle θ, label each side opp, adj or hyp. Then write each ratio as a fraction and as a decimal to 4 decimal places.



$\sin\theta =$ _____

$\cos\theta =$ _____

$\tan\theta =$ _____

8. Three students each drew a right triangle with a 50° angle, then measured the sides.

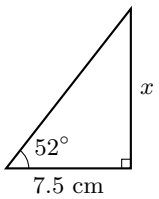
Student	**opp (cm)**	**adj (cm)**	$\frac{\text{opp}}{\text{adj}}$ **(2 decimals)**
Kiara	2.4	2.0	
Luis	3.0	2.5	
Mei	4.2	3.5	

(a) Fill in the last column. What do you notice? Find $\tan 50°$ on your calculator and compare.

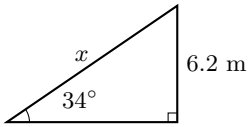
(b) A fourth right triangle has a 50° angle and an adjacent side of 15 cm. Use $\tan 50°$ to find the opposite side.

9. Find each side marked x.

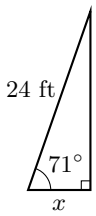
(a)



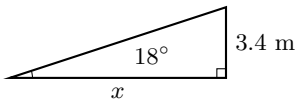
(b)



(c)

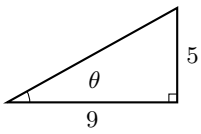


(d)

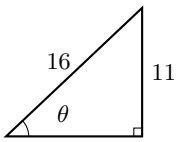


10. Find each angle θ.

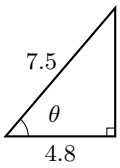
(a)



(b)



(c)



(d) Find the other acute angle in triangle (a).

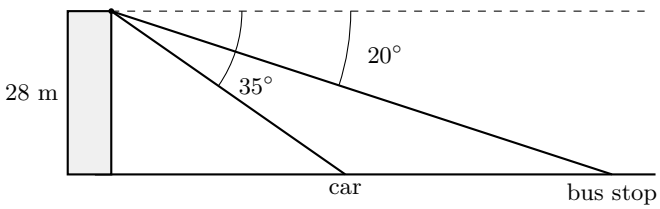
11. A car dealership flies a big advertising balloon on a 70 m rope. The rope is tied to the top of a post 1.1 m tall and makes an angle of elevation of 56°. Assume the rope is straight.

(a) How high is the balloon above the ground?

(b) How far is the post from the point on the ground right under the balloon?

12. In $\triangle DEF$, $\angle F = 90°$, $DF = 9.0$ m and $EF = 13.5$ m. Sketch and label the triangle, then solve it: find DE, $\angle D$ and $\angle E$.

13. From the roof of a 28 m apartment building, Ines looks down at a parked car at an angle of depression of 35°. A bus stop is farther away on the same straight street, at an angle of depression of 20°. How far is the car from the bus stop?



14. A right triangle has a hypotenuse of 12 m and a 40° angle. Tariq wants the side opposite the 40° angle. His work:

$$\sin 40° = \frac{x}{12} \qquad x = 12 \div \sin 40° \approx 18.7 \text{ m}$$

(a) How can you tell his answer is not reasonable, without redoing it?

(b) Find and fix his error.

Part B: Multiple Choice

15. $\sqrt{110}$ is between

A. 9 and 10 **B.** 10 and 11 **C.** 54 and 56 **D.** 100 and 121

16. For an acute angle θ in a right triangle, $\tan\theta$ is

A. $\frac{\text{opp}}{\text{hyp}}$ **B.** $\frac{\text{adj}}{\text{hyp}}$ **C.** $\frac{\text{opp}}{\text{adj}}$ **D.** $\frac{\text{adj}}{\text{opp}}$

17. A right triangle has legs of 5 m and 7 m. To the nearest tenth, its hypotenuse is

A. 12.0 m **B.** 8.6 m **C.** 4.9 m **D.** 74.0 m

18. To check that a corner of a concrete form is square, a builder measures 1.5 m along one side and 2 m along the other. The diagonal between the two marks should be

A. 3.5 m **B.** 2.5 m **C.** 3.0 m **D.** 1.8 m

19. If $\sin 25° = \dfrac{x}{14}$, then to the nearest tenth x is

A. 5.9 **B.** 33.1 **C.** 12.7 **D.** 6.5

20. $\tan\theta = 0.8$. To the nearest degree, θ is

A. 53° **B.** 37° **C.** 51° **D.** 39°

21. From the top of a lookout, the angle of depression to a canoe is 14°. The angle of elevation from the canoe to the top of the lookout is

A. 14° **B.** 76° **C.** 104° **D.** 166°

22. Which number can **not** be the sine of an acute angle?

A. 0.25 **B.** 0.99 **C.** 1.25 **D.** 0.5

Part C: Numerical Response

23. Numerical Response A right triangle has legs of 20 cm and 21 cm. The length of the hypotenuse is _____ cm.

24. Numerical Response A shed roof rises 2.5 m over a horizontal run of 6 m. To the nearest degree, the angle of the roof is _____°.

25. Numerical Response A guy wire 30 m long runs from the top of a tower to the ground and makes an angle of 64° with the ground. To the nearest tenth of a metre, the tower is _____ m tall.

26. Numerical Response A straight ski run is 850 m long and slopes down at 21° from the horizontal. To the nearest metre, the horizontal distance it covers is _____ m.
